

NPTEL Course on
"Computer System Optimization for ML"

Lecture #51

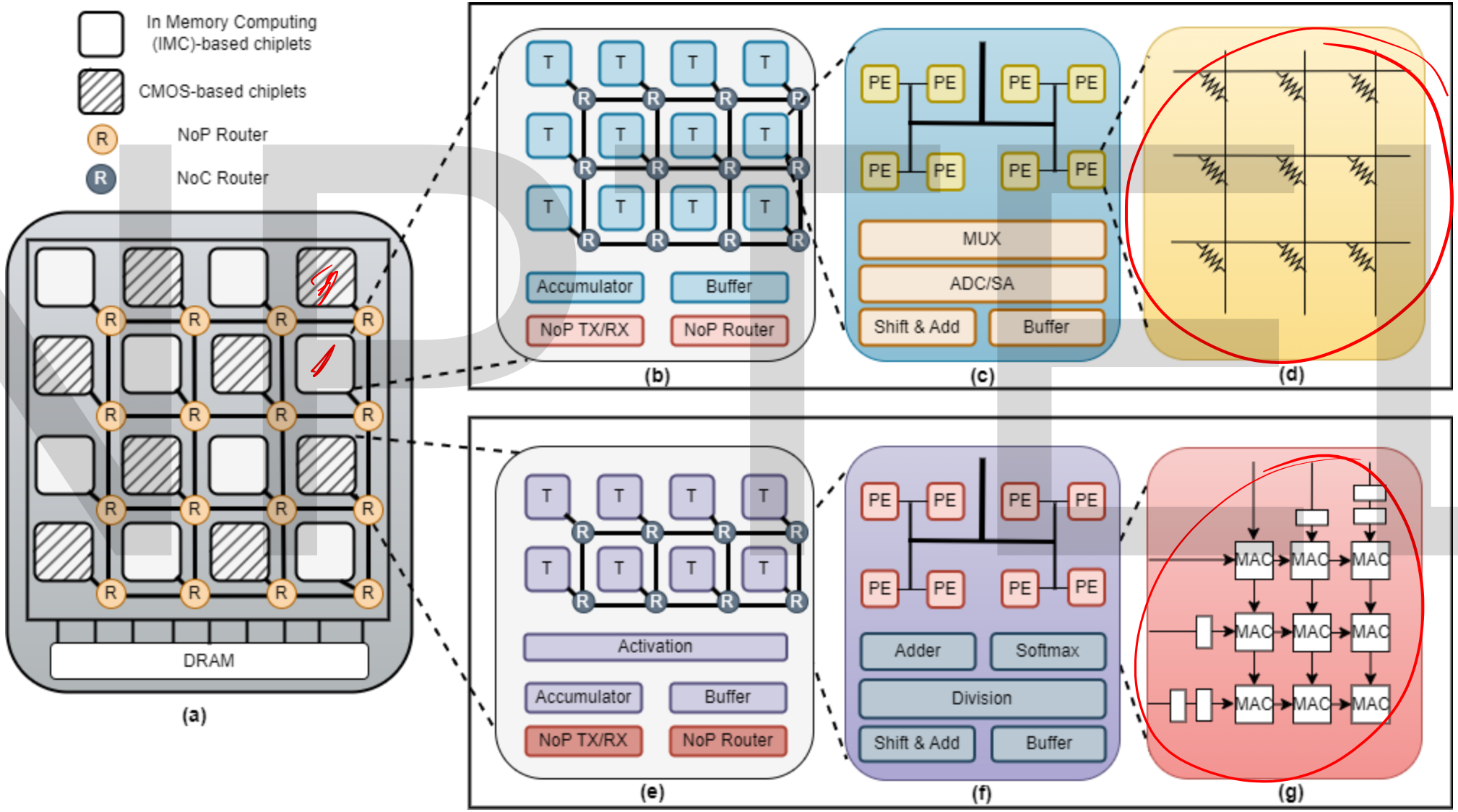
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Agenda

- Chiplet-based Heterogeneous Accelerator for LLMs (contd.)
- Performance Characterization

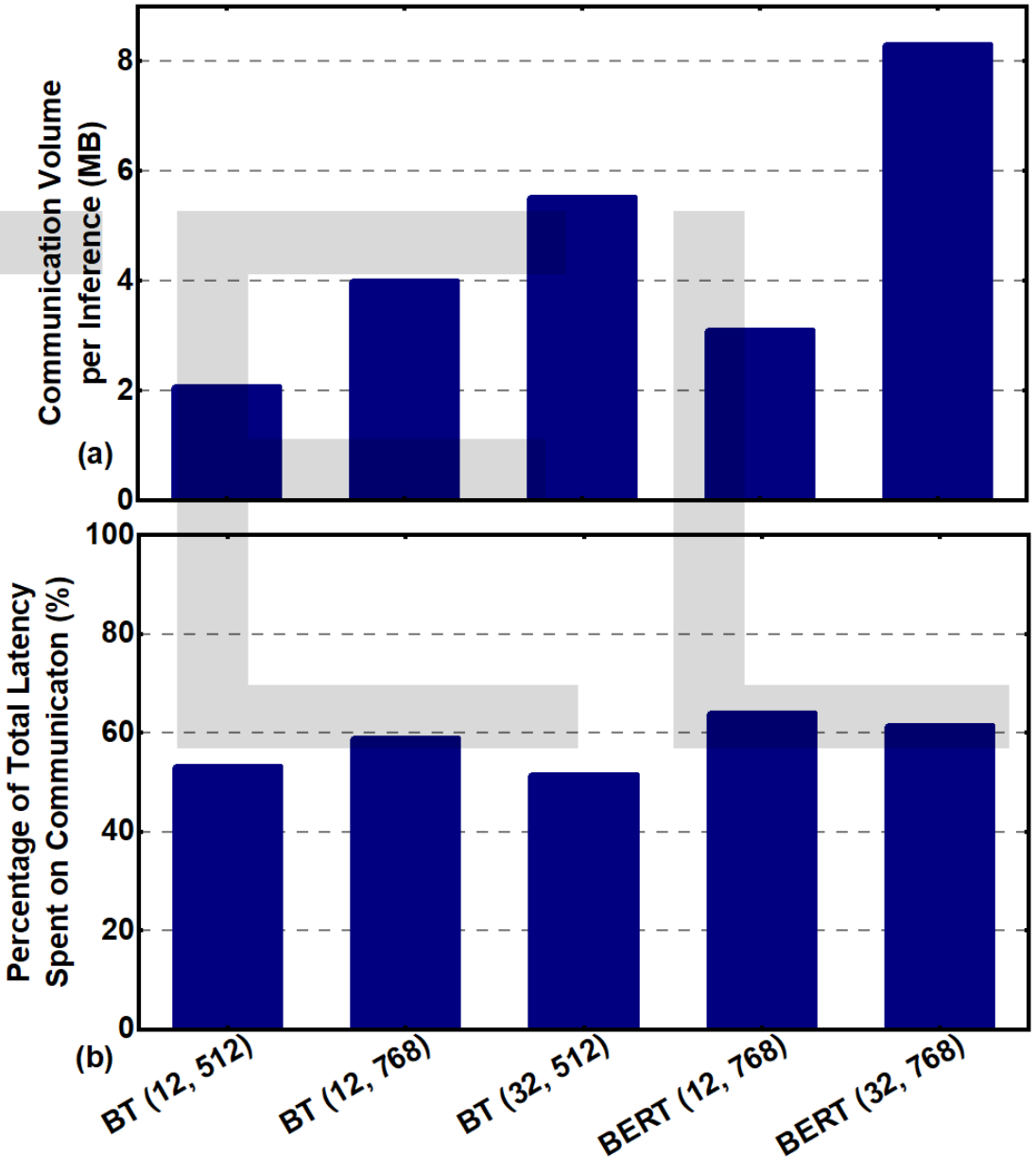
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Top-level Architecture

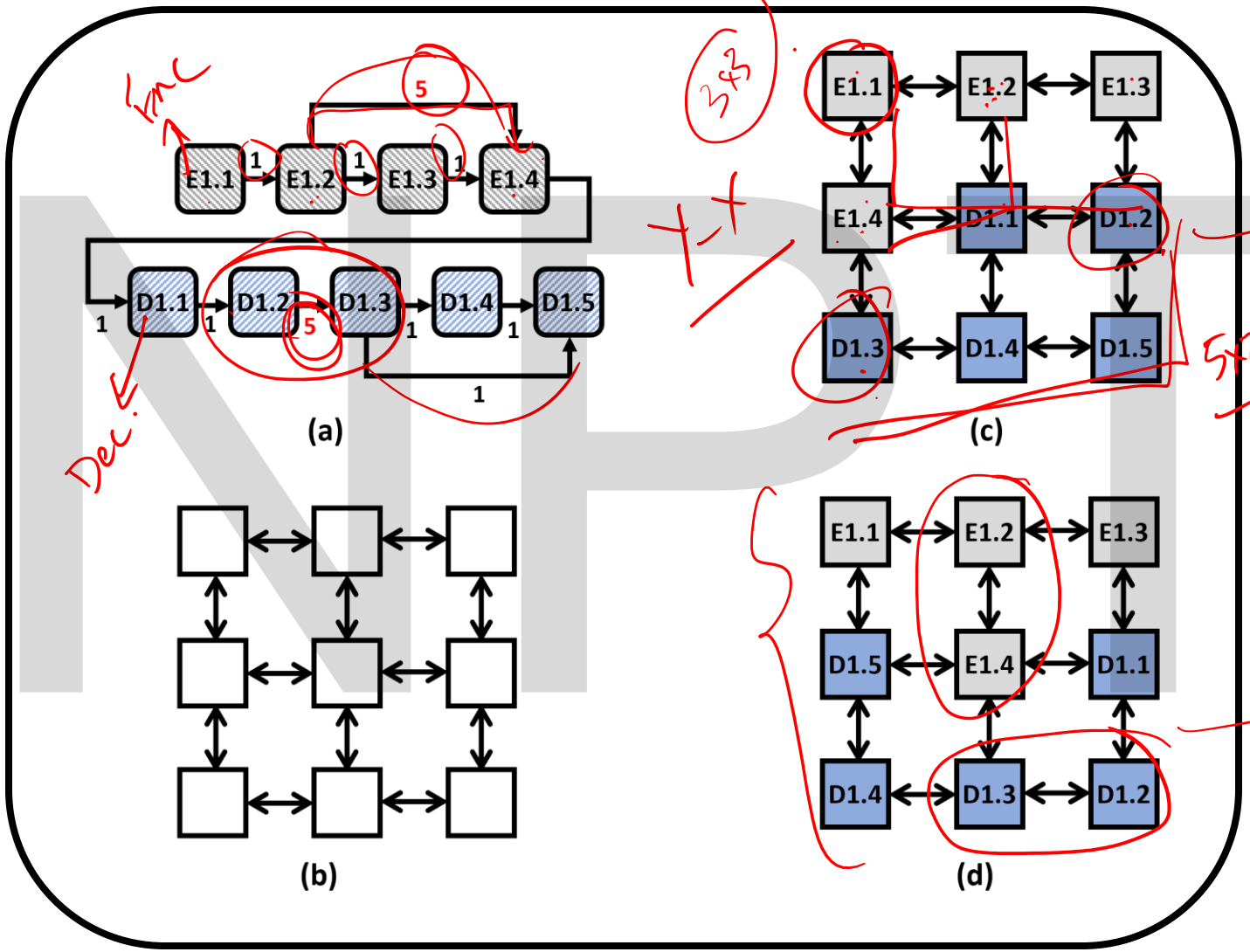


Communication between LLM Phases

- Different phases communicate between each other
- Up to 8 MB of communication volume per inference of LLM
- 50%-65% of total latency is spent on communication
- Communication is the major bottleneck of the performance
  - Mapping is crucial



An Illustrative Example of Mapping

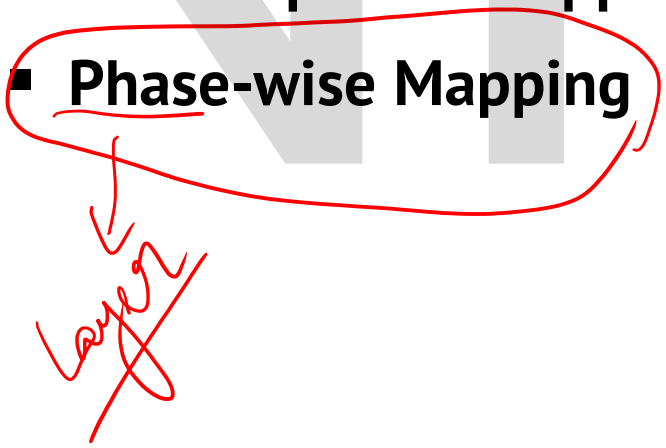


Communication cost = volume × Manhattan distance

s,d	v	M^1	$v.M^1$	M^2	$v.M^2$
E1.1, E1.2	1	1	1	1	1
E1.2, E1.3	1	1	1	1	1
E1.2, E1.4	5	2	10	1	5
E1.3, E1.4	1	3	3	2	2
E1.4, D1.1	1	1	1	1	1
D1.1, D1.2	1	1	1	1	1
D1.2, D1.3	5	3	15	1	5
D1.3, D1.4	1	1	1	1	1
D1.3, D1.5	1	2	2	2	2
D1.4, D1.5	1	1	1	1	1
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Proposal for Mapping

- Scalable technique to map any number of computing phases on to chiplets
  - Objective is to reduce communication cost
- The cost metric is an indicator of congestion
- A combination of branch-and-bound and sequential mapping technique
  - Branch-and-bound is compute intensive if number of combinations are more
  - Sub-optimal mapping through sequential mapping technique
- Phase-wise Mapping

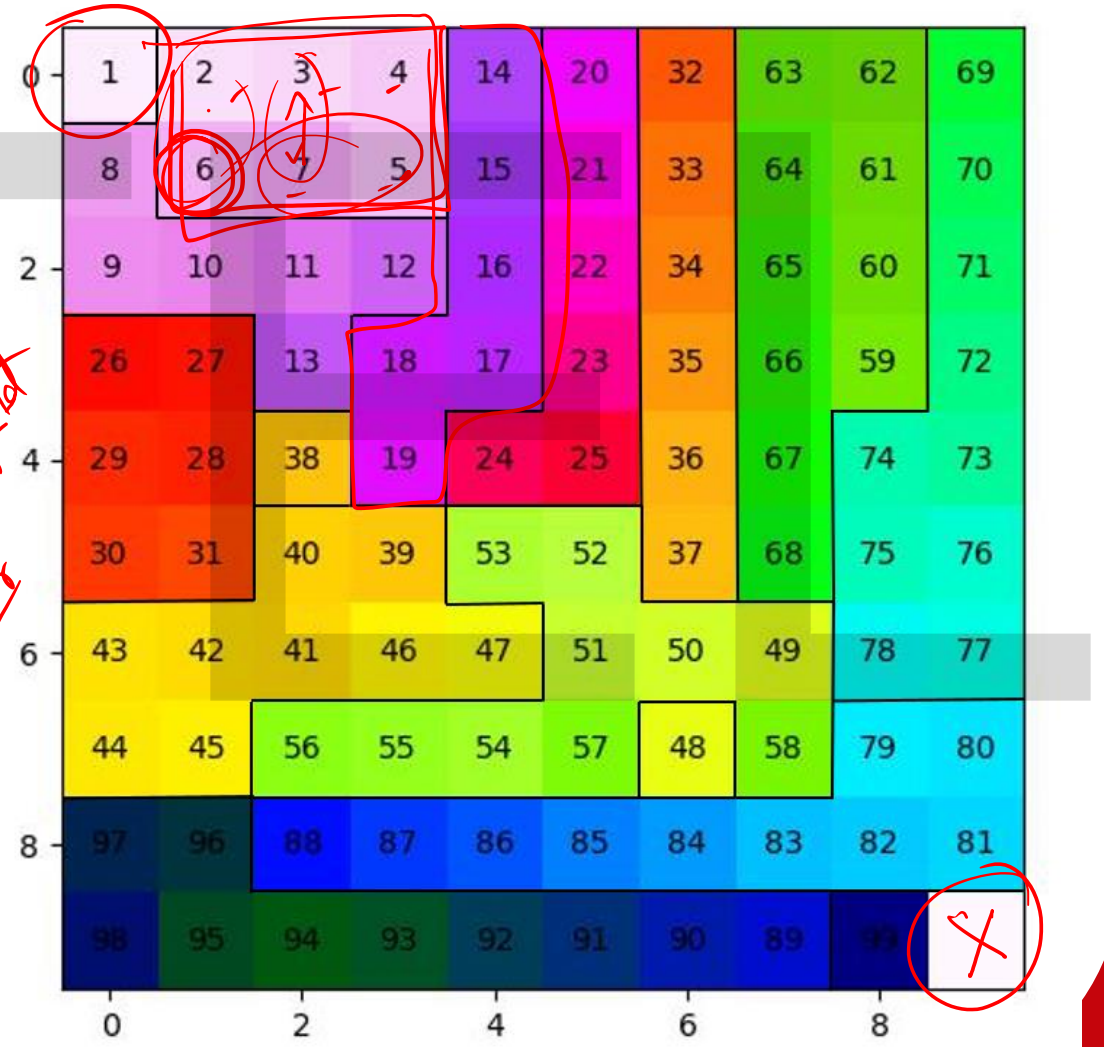




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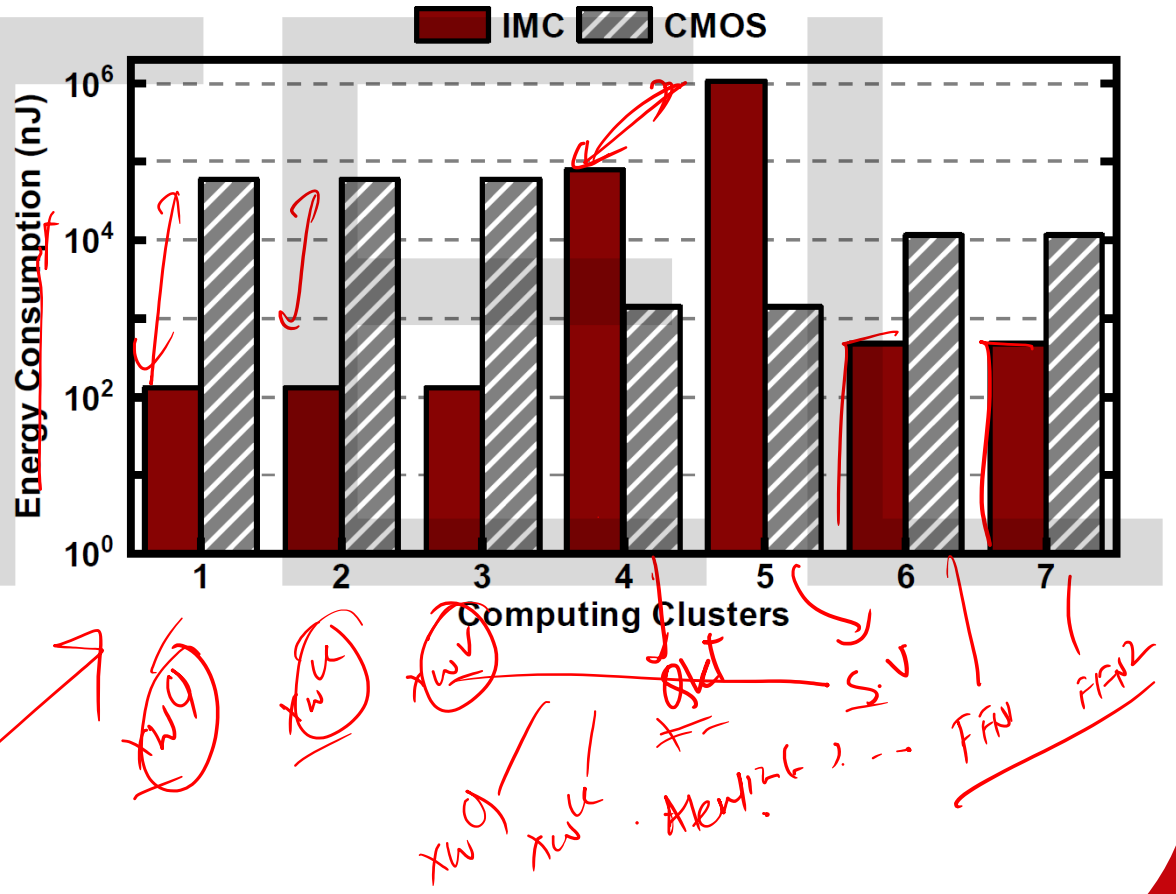
Experimental Evaluation: Example of Mapping

- **Evaluation**
  - High level simulation for IMC chiplets
  - Synthesized the RTL for CMOS chiplets
- **Representative mapping is shown in the figure**
  - Each bounded box is a phase (encoder/decoder)



Comparison between IMC and CMOS

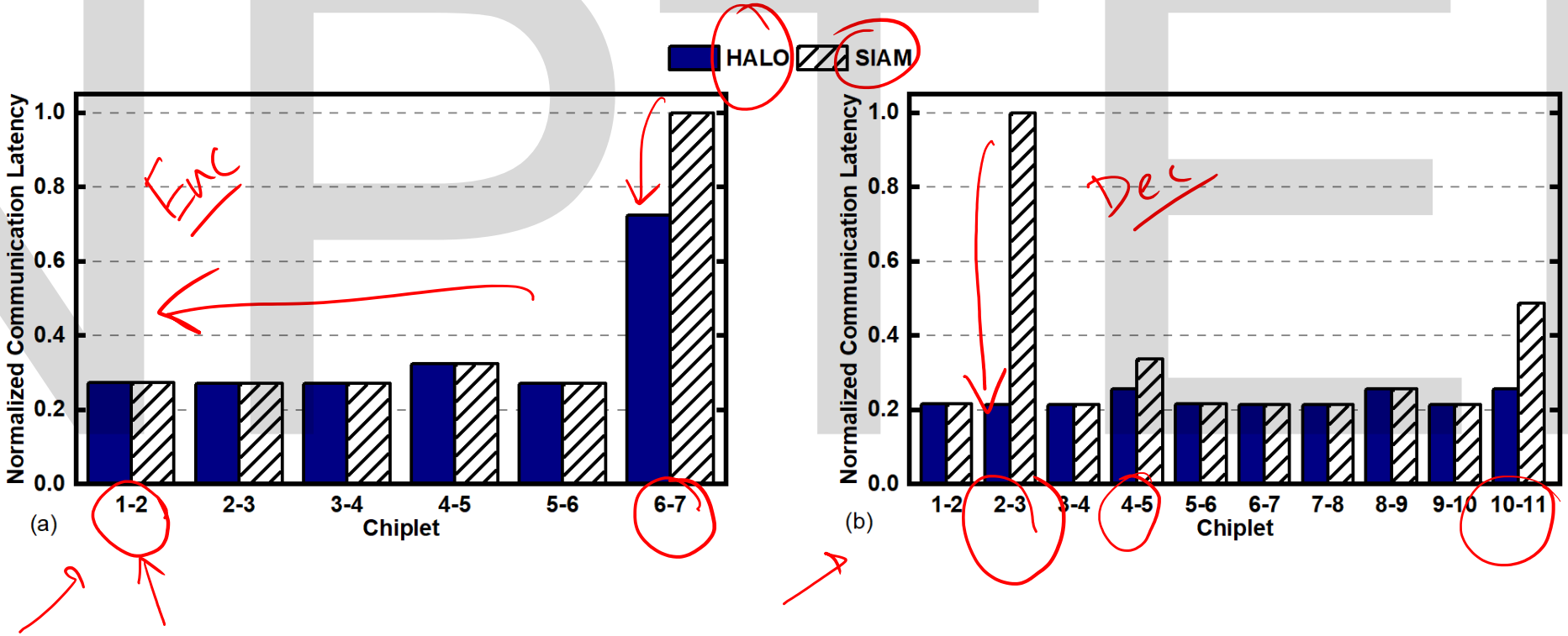
- **Evaluation**
  - High level simulation for IMC chiplets
  - Synthesized the RTL for CMOS chiplets
- **Comparison of IMC vs CMOS**
  - IMC is energy-efficient for weight stationary multiplications
  - CMOS is energy-efficient where both inputs are dynamic



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Comparison in Communication Performance

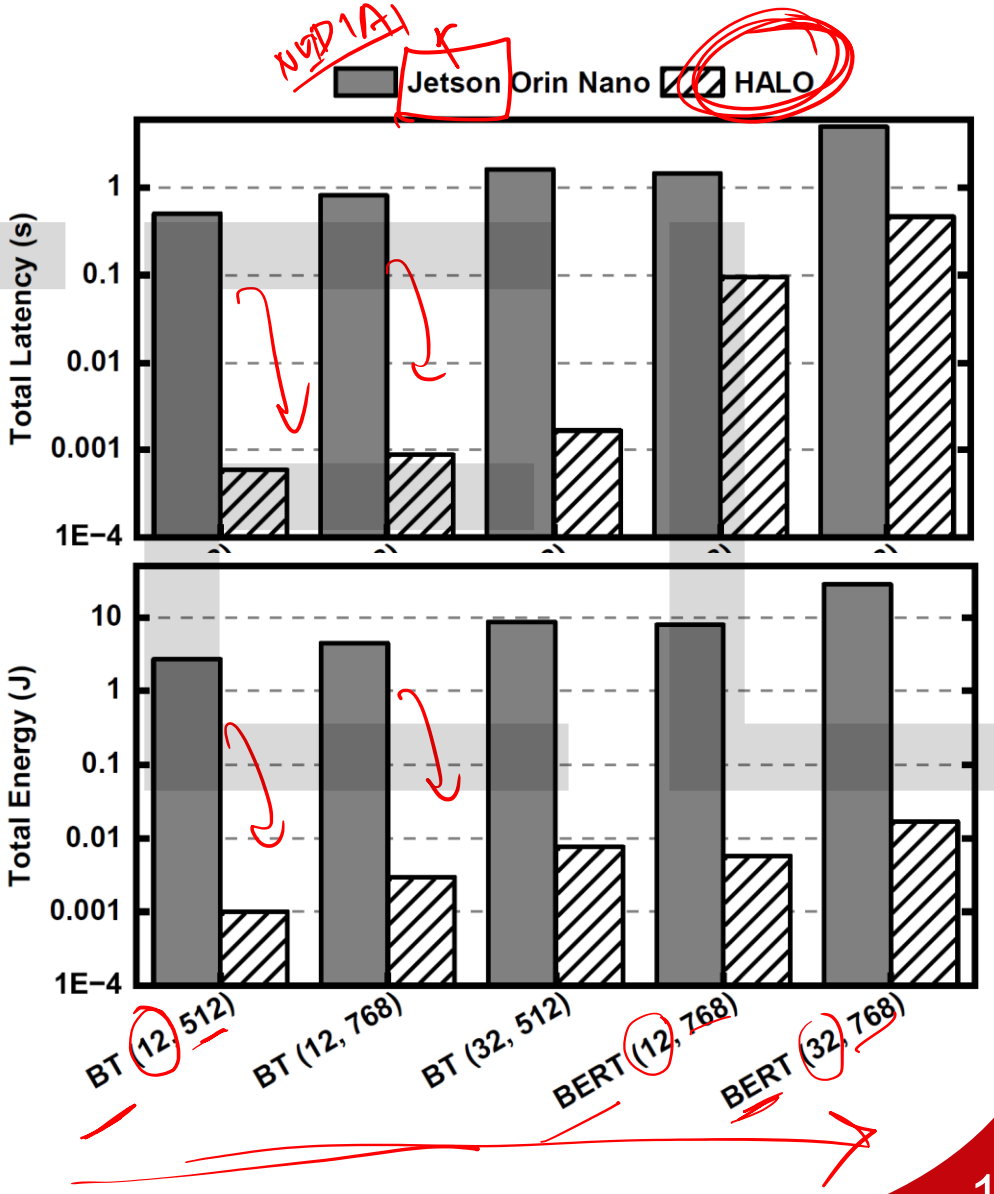
- Chiplet-wise communication performance is shown
- SIAM [6] is the state-of-the-art 2.5D-based ML accelerator



[6] Krishnan, Gokul, et al. "SIAM: Chiplet-based scalable in-memory acceleration with mesh for deep neural networks." *ACM Transactions on Embedded Computing Systems (TECS)* 20.5s (2021): 1-24.

Experimental Evaluation [7]

- **Evaluation**
  - High level simulation for IMC chiplets
  - Synthesized the RTL for CMOS chiplets
- **Comparison with respect to NVIDIA Jetson**
  - 972× improvement in latency
  - 1600× improvement in energy



[7] Jaiswal, Abhi, et al. "HALO: Communication-aware Heterogeneous 2.5 D System for Energy-efficient LLM Execution at Edge." *IEEE Journal on Emerging and Selected Topics in Circuits and Systems* (2024).

Spotlight article in Q3'24



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)